

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyse DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Concept Mapping (Medium) Assessment

Tool Weightage: 20%

QUESTIONS		
Q.No	Question	Bloom's Level
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand
6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand

7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand
8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.	BL1 – Remember

Code of Conduct:

I R, INDHU (192524332)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
R. INDHU

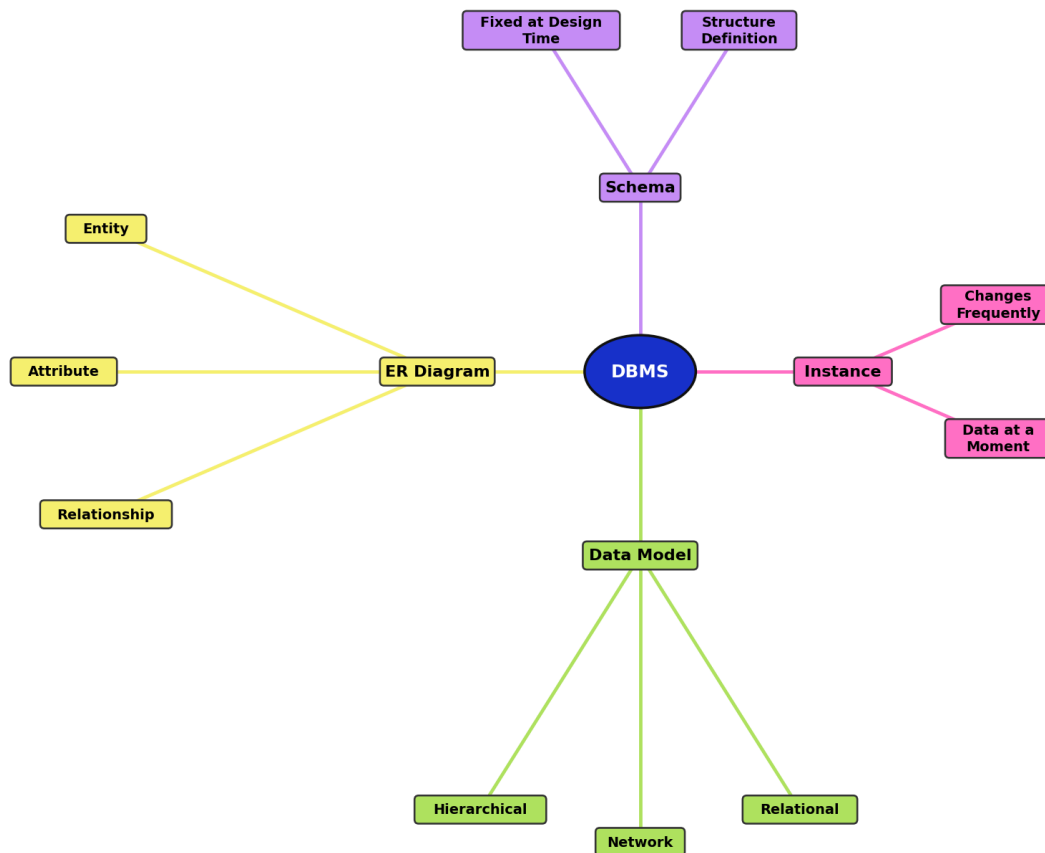
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Concept Map 1: DBMS Core Terminology

Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.

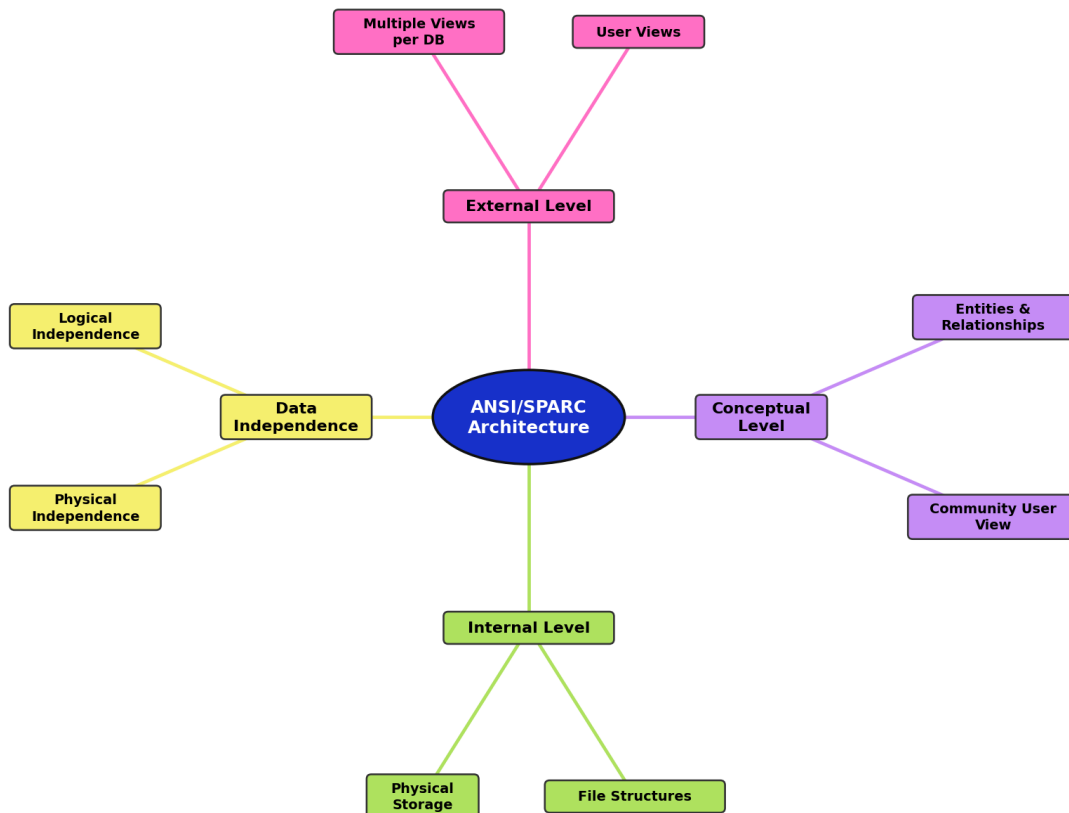


Key Points

- A DBMS is the software layer that manages a database; it is described by a Schema (structure) and holds an Instance (the actual data at a point in time).
- The Schema is defined once at design time, while the Instance changes every time data is inserted, updated, or deleted.
- A Data Model (Hierarchical, Network, or Relational) is the set of rules used to define how the schema is structured.
- An ER Diagram is the design tool used to plan a schema, built from Entities, Attributes, and Relationships.
- Together these terms show the path from conceptual design (ER Diagram) to a working database (Schema + Instance).

Concept Map 2: ANSI/SPARC Three-Level Architecture

Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.

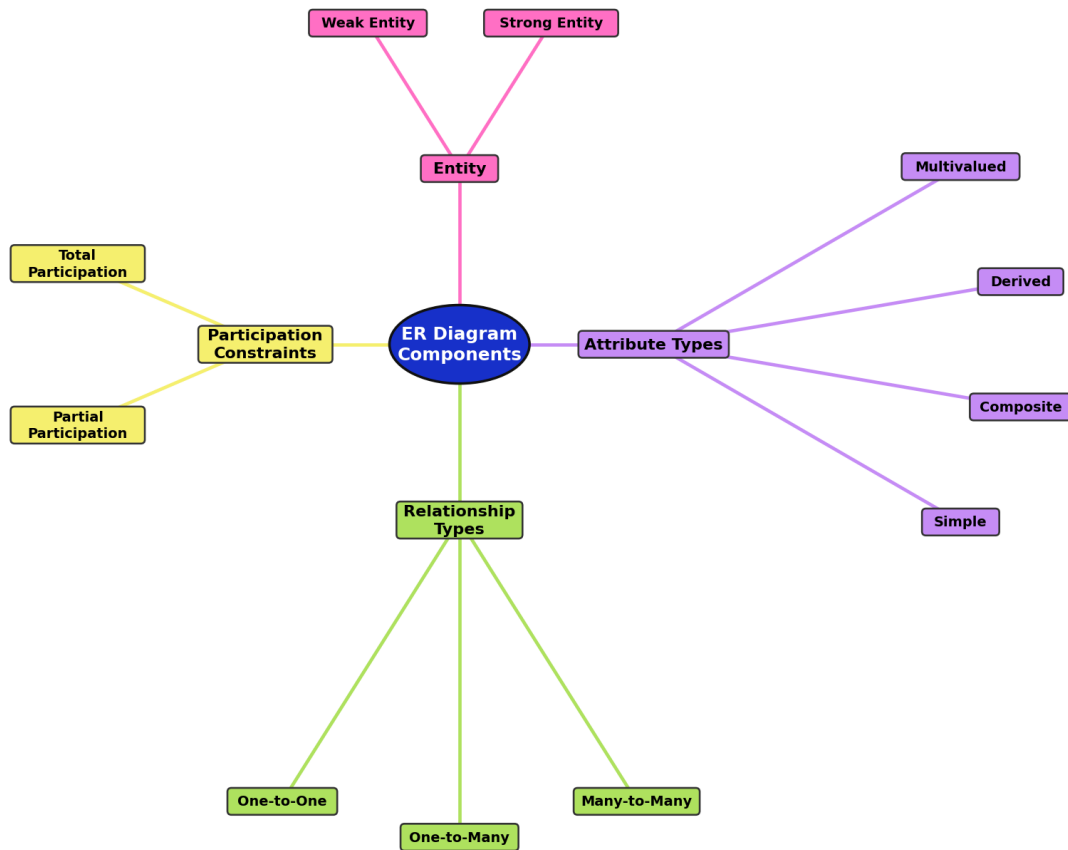


Key Points

- The External Level defines individual user views — different users can see different subsets of the same database.
- The Conceptual Level describes the community (whole-database) view — all entities, relationships, and constraints.
- The Internal Level describes how data is physically stored — file organization, indexing, and storage
- Logical Data Independence lets the conceptual schema change without affecting external views.
- Physical Data Independence lets the internal schema change (e.g. storage method) without affecting the conceptual schema.

Concept Map 3: ER Diagram Components

Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.

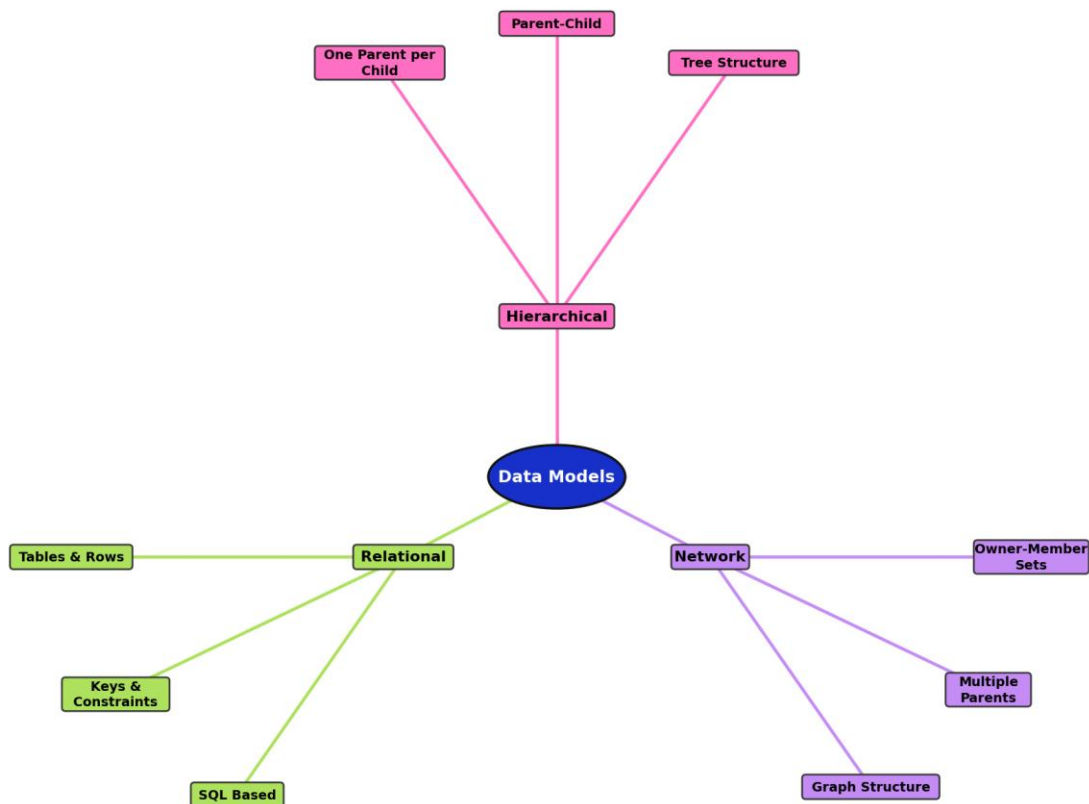


Key Points

- A Strong Entity has its own primary key; a Weak Entity depends on another entity for identification.
- Attributes can be Simple, Composite, Derived, or Multivalued depending on how they store values.
- Relationship Types describe cardinality between entities: One-to-One, One-to-Many, or Many-to-Many.
- Participation Constraints define whether every entity instance must take part in a relationship (Total) or only some do (Partial).
- These components together form the building blocks used to translate a real-world scenario into an ER diagram.

Concept Map 5: Comparing Data Models

Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.

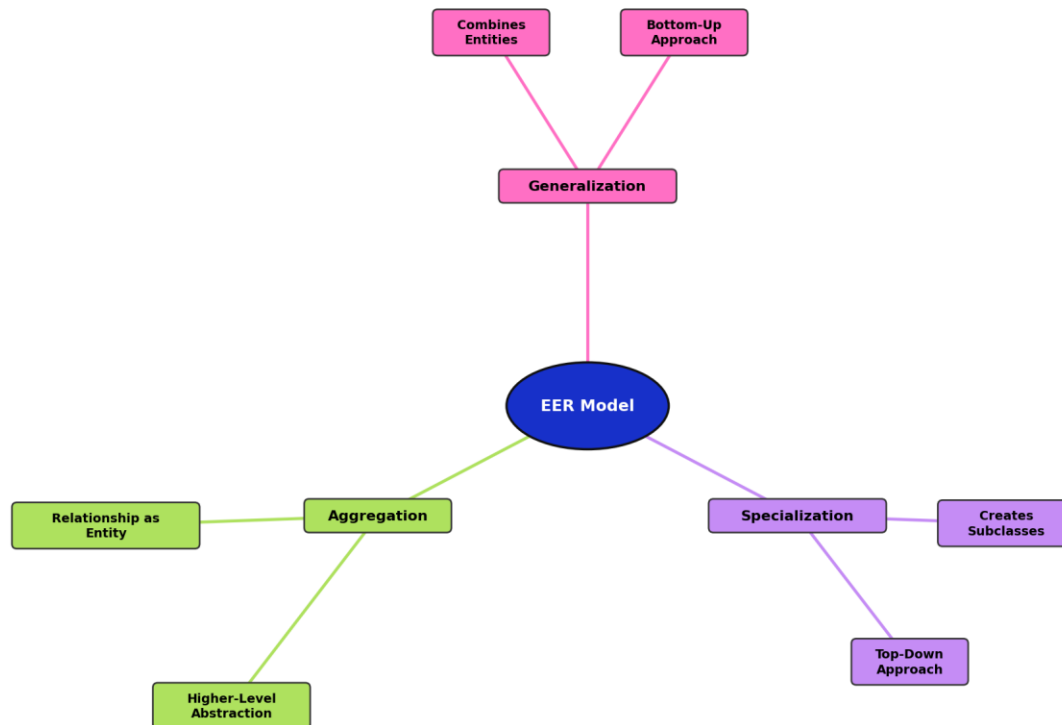


Key Points

- The Hierarchical model organizes data as a tree, where each child record has exactly one parent.
- The Network model organizes data as a graph, allowing a child (member) to have multiple parent (owner) records.
- The Relational model organizes data as tables (relations) of rows and columns, linked through keys.
- Relational databases are queried using SQL, while hierarchical and network models rely on navigational access.
- Relational is the most flexible and widely used today because it separates logical structure from physical storage.

Concept Map 6: The EER Model

Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.

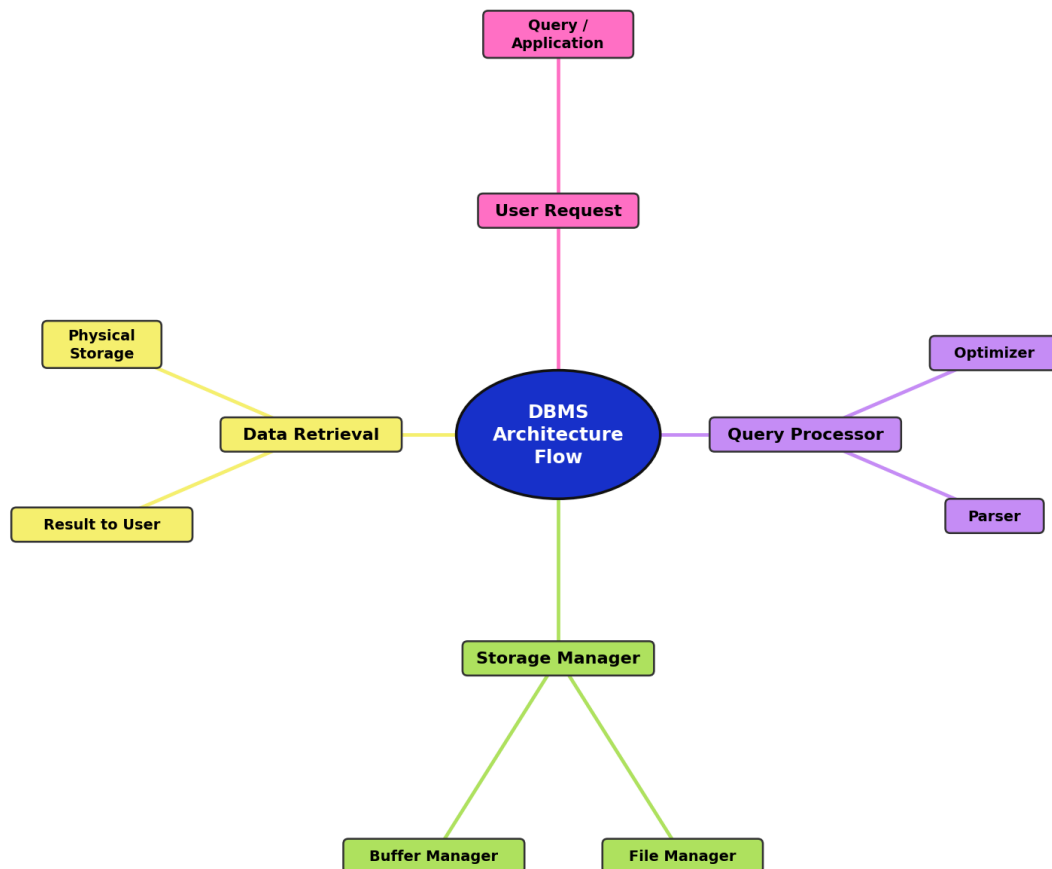


Key Points

- Generalization is a bottom-up process that combines several similar entities into one general superclass.
- Specialization is a top-down process that divides a general entity into more specific subclasses.
- Aggregation treats a relationship itself as a higher-level entity so it can take part in further relationships.
- Together, generalization and specialization form class/subclass hierarchies with attribute and relationship inheritance.
- The EER model extends the basic ER model to capture these more complex, real-world structures.

Concept Map 7: DBMS Software Architecture Flow

Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.

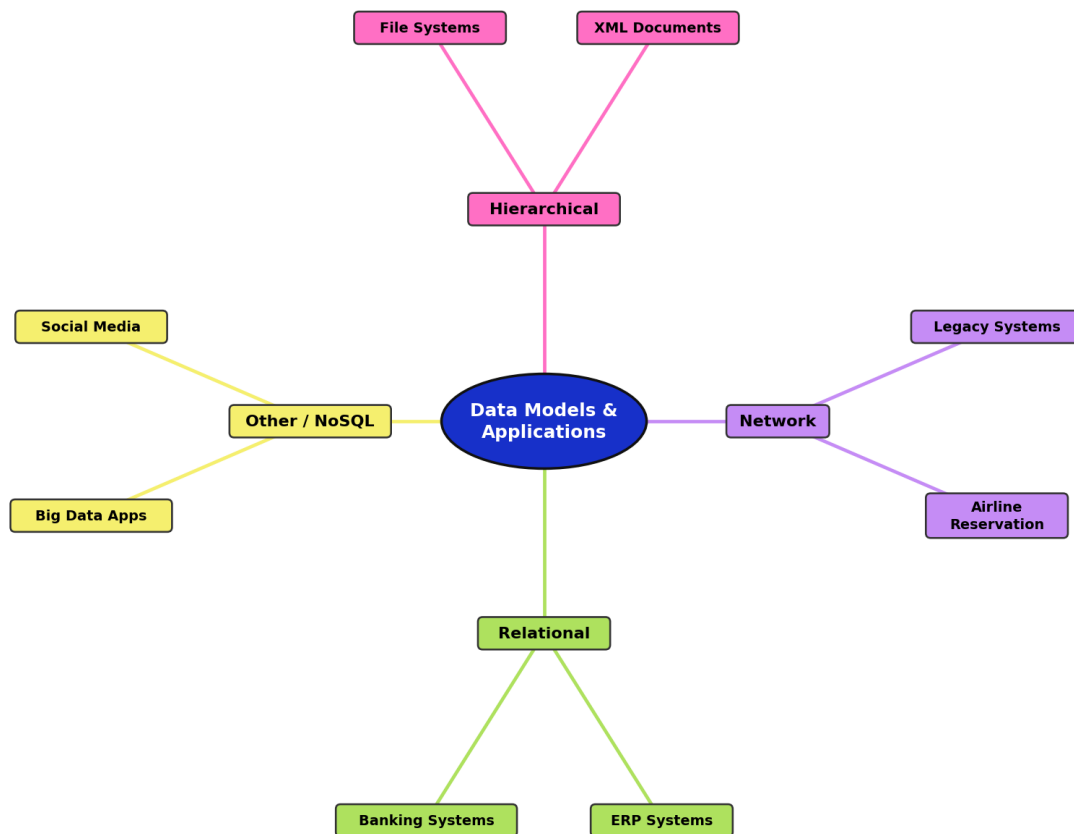


Key Points

- A User Request originates from an application program or an ad-hoc query.
- The Query Processor parses the request and the Optimizer chooses the most efficient execution plan.
- The Storage Manager coordinates the Buffer Manager (in-memory pages) and File Manager (data on disk).
- Data Retrieval fetches the required records from physical storage and returns the result set to the user.
- Each stage hands off to the next, forming a pipeline from request to result.

Concept Map 8: Data Models Mapped to Real-World Applications

Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.

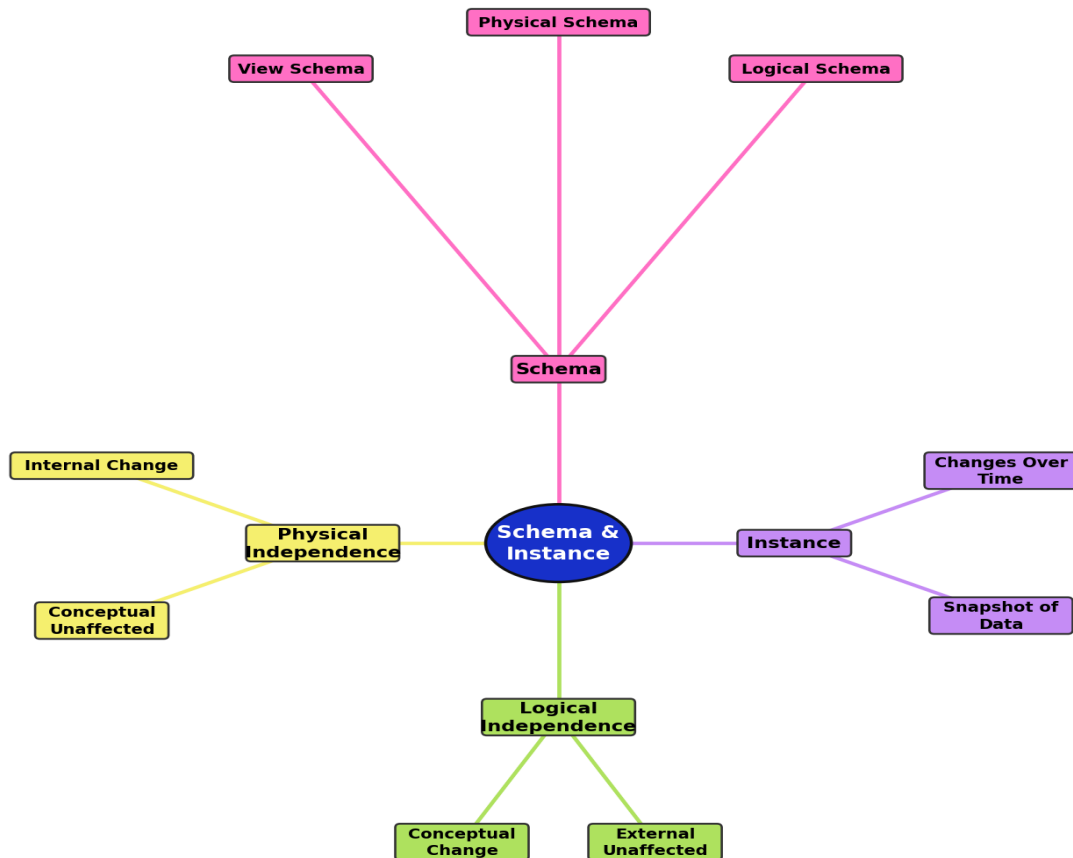


Key Points

- Hierarchical models suit XML documents and simple file systems where a strict parent-child order fits naturally.
- Network models historically suited airline reservation and legacy systems needing many-to-many navigational links.
- Relational models suit banking and ERP systems that need strong consistency, constraints, and SQL-based queries.
- Newer NoSQL-style models suit social media and big-data applications needing flexible, high-volume, semi-structured data.
- Choosing a model depends on data structure, query pattern, consistency needs, and scale of the application.

Concept Map 9: Schemas, Instances & Data Independence

Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.

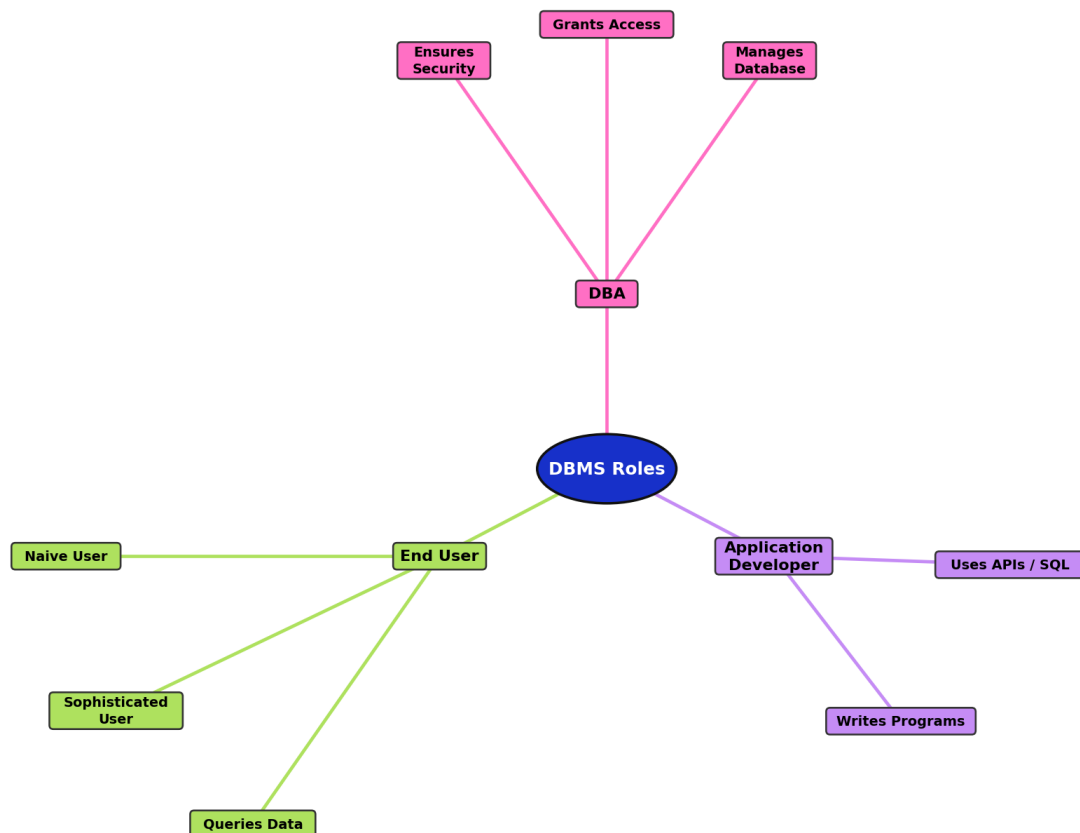


Key Points

- A Schema exists at three levels — logical (conceptual), physical (internal), and view (external) — each describing structure, not data.
- An Instance is the actual data present at a given moment, and it changes far more often than the schema.
- Logical Independence means the conceptual schema can change without forcing changes to external views.
- Physical Independence means the internal (storage) schema can change without affecting the conceptual schema.
- This separation of schema levels is what allows a database to evolve without breaking existing applications.

Concept Map 10: Roles & Responsibilities in a DBMS Environment

Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.



Key Points

- The DBA (Database Administrator) manages the database, grants user access, and ensures overall security.
- The Application Developer writes programs that interact with the database through APIs or embedded SQL.
- End Users are split into Naive Users (use pre-built forms/menus) and Sophisticated Users (write their own queries).
- The DBA's decisions on schema and access directly shape what developers can build and what end users can see.
- Clear separation of these roles keeps the database secure, maintainable, and usable across an organization.